When one service needs to communicate with another, it must be provided with information about where to find it, such as an IP address or a DNS name. Let's consider two applications: Alpha App and Beta App. Figure 7.6 shows how the communication between the two would happen if there were only one Beta App instance.

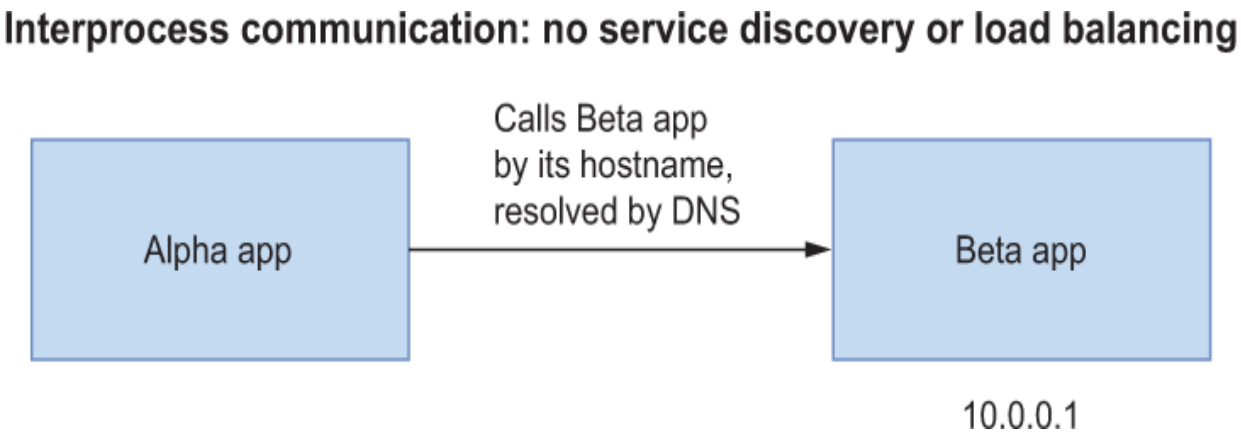


Figure 7.6 If there were only one Beta App instance, the interprocess communication between Alpha App and Beta App would be based on a DNS name resolving to the IP address of Beta App.

In the scenario illustrated in figure 7.6, we say that Alpha App is *upstream* and Beta App is *downstream*. Furthermore, Beta App is a *backing service* with respect to Alpha App. There's only one instance of Beta App running, so the DNS name gets resolved to its IP address.

In the cloud, you'll probably want to have multiple instances of a service running, and each service instance will have its own IP address. Unlike physical machines or long-running virtual machines, a service instance will not live long in the cloud. Application instances are disposable—they can be removed or replaced for different reasons, such as when they are not responsive anymore. You can even enable the auto-scaling feature to automatically scale your application in and out, depending on the workload.